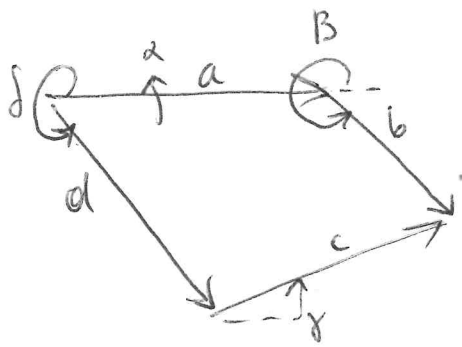


ES 1)



|               |          |         |          |     |
|---------------|----------|---------|----------|-----|
| $\cos \theta$ | $a$      | $b$     | $c$      | $d$ |
| $\text{VAR}$  | $\alpha$ | $\beta$ | $\gamma$ |     |

imposed:  $\dot{\alpha} = 1 \text{ rad/s}$ ,  $\ddot{\alpha} = 1 \text{ rad/s}^2$

$$a e^{i\alpha} + b e^{i\beta} = c e^{i\gamma} + d e^{i\delta}$$

$$\begin{cases} a \cos \alpha + b \cos \beta = c \cos \gamma + d \cos \delta \\ a \sin \alpha + b \sin \beta = c \sin \gamma + d \sin \delta \end{cases}$$

Velocity

$$\begin{cases} -a \dot{\alpha} \sin \alpha - b \dot{\beta} \sin \beta = -c \dot{\gamma} \sin \gamma \\ a \dot{\alpha} \cos \alpha + b \dot{\beta} \cos \beta = c \dot{\gamma} \cos \gamma \end{cases} \rightarrow \begin{bmatrix} b \sin \beta & -c \sin \gamma \\ -b \cos \beta & c \cos \gamma \end{bmatrix} \begin{bmatrix} \dot{\beta} \\ \dot{\gamma} \end{bmatrix} = \begin{bmatrix} -a \dot{\alpha} \sin \alpha \\ a \dot{\alpha} \cos \alpha \end{bmatrix}$$

$$\begin{cases} \dot{\beta} = -1,03 \text{ rad/s} \\ \dot{\gamma} = 0,97 \text{ rad/s} \end{cases} \rightarrow \vec{v}_n = \dot{\gamma} R \quad \vec{j} = 0,39 \frac{m}{s} \vec{j}$$

acceleration

$$\begin{cases} -a \ddot{\alpha} \sin \alpha - a \dot{\alpha}^2 \cos \alpha - b \ddot{\beta} \sin \beta - b \dot{\beta}^2 \cos \beta = -c \ddot{\gamma} \sin \gamma - c \dot{\gamma}^2 \cos \gamma \\ a \ddot{\alpha} \cos \alpha - a \dot{\alpha}^2 \sin \alpha + b \ddot{\beta} \cos \beta - b \dot{\beta}^2 \sin \beta = c \ddot{\gamma} \cos \gamma - c \dot{\gamma}^2 \sin \gamma \end{cases}$$

$$\rightarrow \begin{bmatrix} b \sin \beta & -c \sin \gamma \\ -b \cos \beta & c \cos \gamma \end{bmatrix} \begin{bmatrix} \ddot{\beta} \\ \ddot{\gamma} \end{bmatrix} = \begin{bmatrix} -a \ddot{\alpha} \sin \alpha - a \dot{\alpha}^2 \cos \alpha - b \dot{\beta}^2 \cos \beta + c \dot{\gamma}^2 \cos \gamma \\ a \ddot{\alpha} \cos \alpha - a \dot{\alpha}^2 \sin \alpha - b \dot{\beta}^2 \sin \beta + c \dot{\gamma}^2 \sin \gamma \end{bmatrix}$$

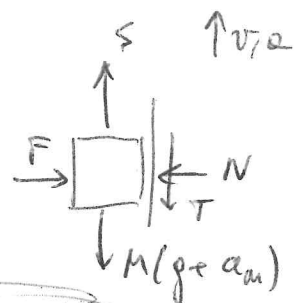
$$\begin{cases} \ddot{\beta} = -0,43 \text{ rad/s}^2 \\ \ddot{\gamma} = 2,43 \text{ rad/s}^2 \end{cases} \rightarrow \vec{a}_n = \ddot{\gamma} R \quad \vec{j} = 0,97 \frac{m}{s^2} \vec{j}$$

$$\begin{aligned} \frac{dE_c}{dt} &= m v_n a_n + J_c \ddot{\alpha} + J_d \ddot{\gamma} + M v_n a_n = \\ &= \left( m \frac{a^2}{4} + J_c \right) \ddot{\alpha} + (I R^2 + J_d) \ddot{\gamma} = 4,48 \text{ W} \end{aligned}$$

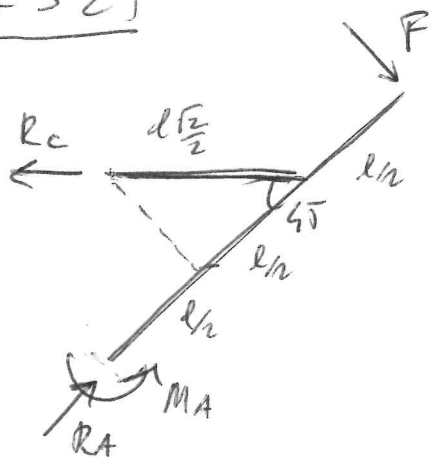
$$\begin{aligned} \Sigma W_{\text{ext}} &= m \vec{g} \cdot \vec{v}_n + C_m \dot{\alpha} + M \vec{g} \cdot \vec{v}_n - f_d |N v_n| = \\ &= -m g \dot{\alpha} \frac{a}{2} + C_m \dot{\alpha} - M g \dot{\gamma} R - f_d F \dot{\gamma} R \end{aligned}$$

$$\rightarrow C_m = \frac{dE_c/dt + m g \dot{\alpha} \frac{a}{2} + M g \dot{\gamma} R + f_d F \dot{\gamma} R}{\dot{\alpha}} = 36,13 \text{ Nm}$$

$$S = M(g + a_n) + f_d F = 31,78 \text{ N}$$



ES 2



$$\sum F_y = 0 \quad R_A \sin 45^\circ - F \sin 45^\circ = 0$$

$$R_A = F$$

$$\sum F_x = 0 \quad R_A \cos 45^\circ - R_c + F \cos 45^\circ = 0$$

$$\frac{F\sqrt{2}}{2} - R_c + \frac{F\sqrt{2}}{2} = 0$$

$$R_c = F\sqrt{2}$$

$$\sum M_A = 0 \quad M_A - F \frac{3l}{2} + R_c l \frac{\sqrt{2}}{2} = 0$$

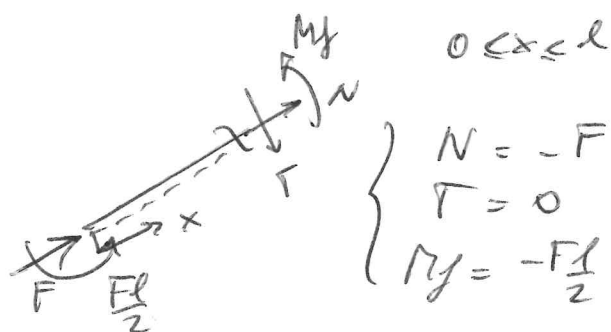
$$M_A - F \frac{3l}{2} + F\sqrt{2} l \frac{\sqrt{2}}{2} = 0$$

$$M_A = \frac{Fl}{2}$$

$$R_c \quad R_D$$

$$R_D = R_c = F\sqrt{2}$$

(←|+|→)

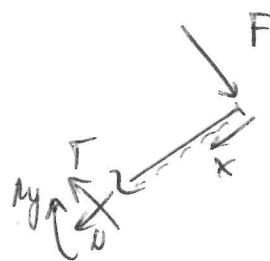


$$0 \leq x \leq l$$

$$N = -F$$

$$T = 0$$

$$M_y = -F \frac{x}{2}$$

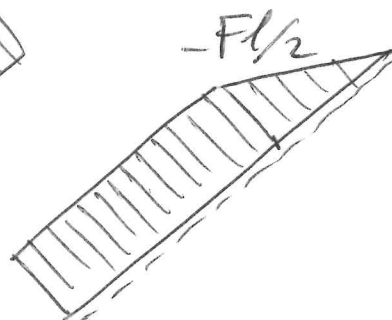
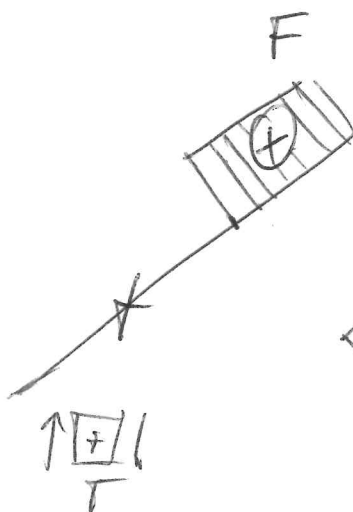
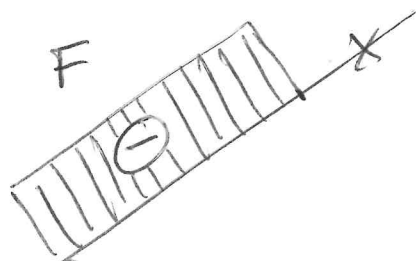


$$0 \leq x \leq \frac{l}{2}$$

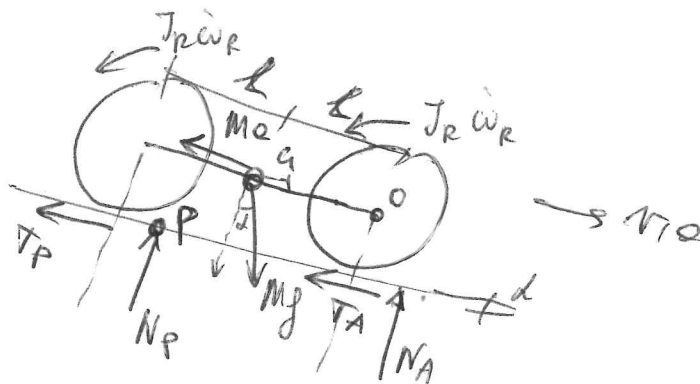
$$N = 0$$

$$T = F$$

$$M_y = -Fx = \begin{cases} 0 \\ -F \frac{x}{2} \end{cases}$$



IES 3



$$\omega_R = \tau \omega_m$$

$$v = \omega_R R = \tau R \omega_m$$



$$W_1 = (C m - J_m \dot{\omega}_m) \omega_m$$

$$\begin{aligned} \Sigma W_{\text{Att}}^{(w)} &= M g v \sin \alpha - N_A f v - N_P f v = M g v \sin \alpha - M g v \cos \alpha f \\ &= M g (\sin \alpha - f \cos \alpha) \tau R \omega_m \\ &= C \dot{\omega}^* \omega_m = (3,4 \text{ Nm}) \omega_m \end{aligned}$$

$$\uparrow N_A + N_P = M g \cos \alpha$$

$$\begin{aligned} \frac{d\bar{L}^{(w)}}{dt} &= M r a + 2 J_A \omega_2 \dot{\omega}_R = (M R^2 + 2 J_A) \tau^2 \omega_m \dot{\omega}_m = \\ &= J \dot{\omega}^* \omega_m \dot{\omega}_m = (0,0488 \text{ kpm}^2) \omega_m \dot{\omega}_m \end{aligned}$$

① Repulse

$$W_2 = \Sigma W_{\text{Att}}^{(w)} = C \dot{\omega}^* \omega_m > 0 \rightarrow \text{Retrógredo}$$

$$W_1 + W_2 \eta_R = 0 \rightarrow C \dot{\omega}^* \omega_m + \eta_R C \dot{\omega}^* \omega_m = 0 \rightarrow C_m = -\frac{1}{\eta_R} C \dot{\omega}^* = (-2,72 \text{ Nm})$$

$$\omega_m = \frac{C_{m0} - C_m \omega_s}{C_{m0}} = 3136 \text{ rpm}$$

$$\textcircled{2} a = 1,5 \text{ m/s}^2 \rightarrow \dot{\omega}_m = \frac{a}{\tau R} = 100 \text{ rad/s}^2$$

$$W_2 = \Sigma W_{\text{Att}}^{(w)} - \frac{d\bar{L}^{(w)}}{dt} = (C \dot{\omega}^* - J \dot{\omega}^* \dot{\omega}_m) \omega_m = (-1,47 \text{ Nm}) \omega_m < 0$$

$$\rightarrow \text{Direto} \rightarrow W_1 \eta_D + W_2 = 0 \rightarrow (C_m - J_m \dot{\omega}_m) \omega_m \eta_D + (C \dot{\omega}^* - J \dot{\omega}^* \dot{\omega}_m) \omega_m = 0$$

$$\rightarrow C_m = J_m \dot{\omega}_m - \frac{(C \dot{\omega}^* - J \dot{\omega}^* \dot{\omega}_m)}{\eta_D} = 51,64 \text{ Nm}$$

③ Adesão anula

$$\Sigma \tau_P^{\text{TOT}} = 0 \quad 2 J_R \dot{\omega}_R + (M a - M g \sin \alpha) R - M g \cos \alpha (L - f R) + N_A 2 L = 0$$

$$\rightarrow N_A = \frac{2 J_R a / R + (M a - M g \sin \alpha) R - M g \cos \alpha (L - f R)}{2 L} = 709,8 \text{ N}$$

$$\Sigma M_O^{(\text{AND})} = 0 \quad J_R \dot{\omega}_R + N_A f R - T_A R = 0$$

$$\rightarrow T_A = \frac{J_R a / R + N_A f R}{R} = 64,2 \text{ N}$$

$$T_{\text{lim}} = f_s N_A = 567,8 \text{ N} > T_A$$

Adesão ✓